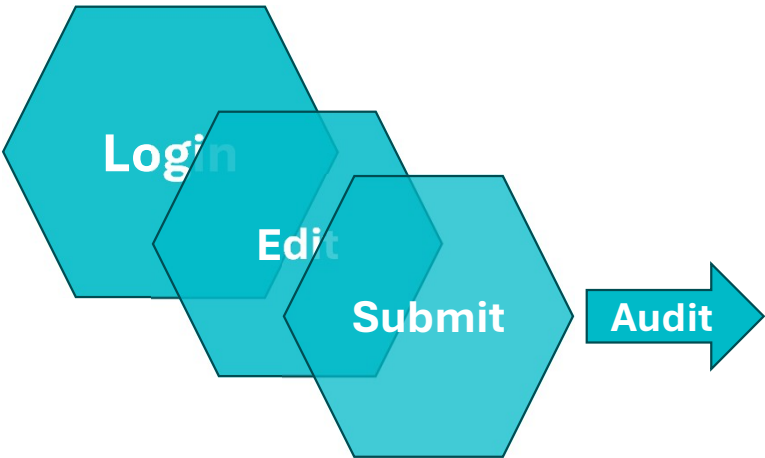


19 December 2025

*SAS Timesheet
Automation
(Week 1)*



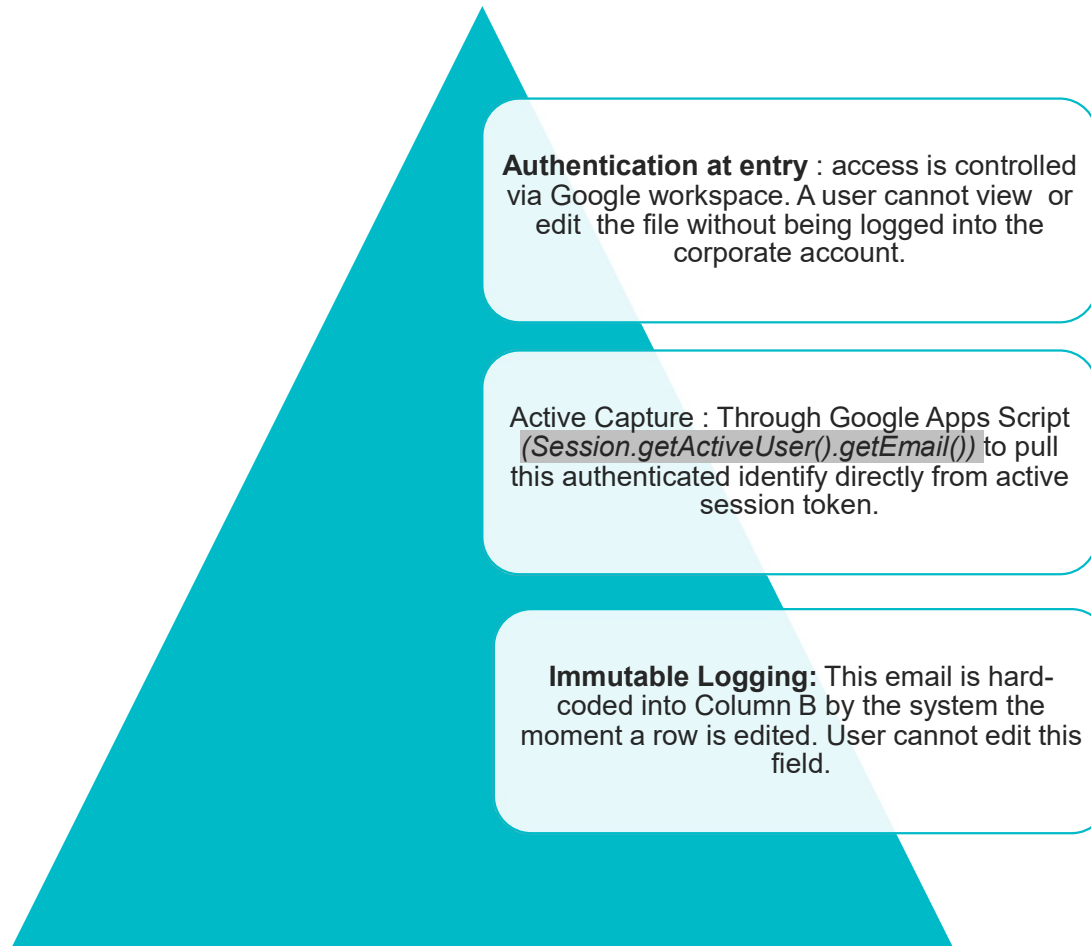
Week 1 Implementation Overview



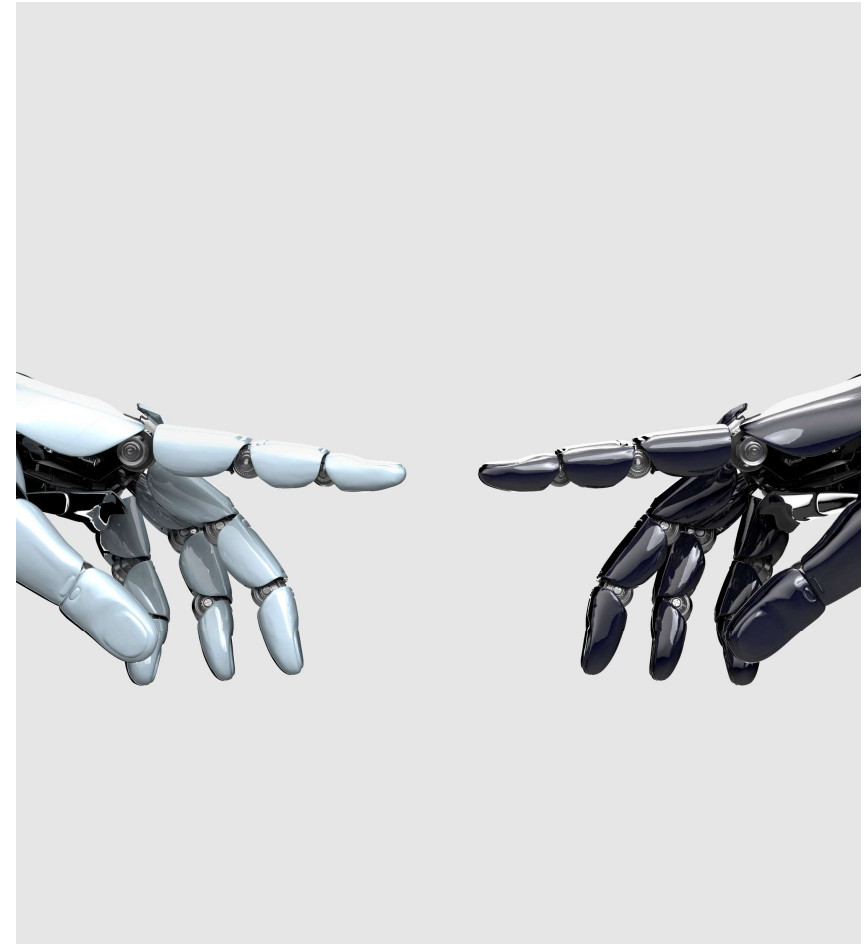
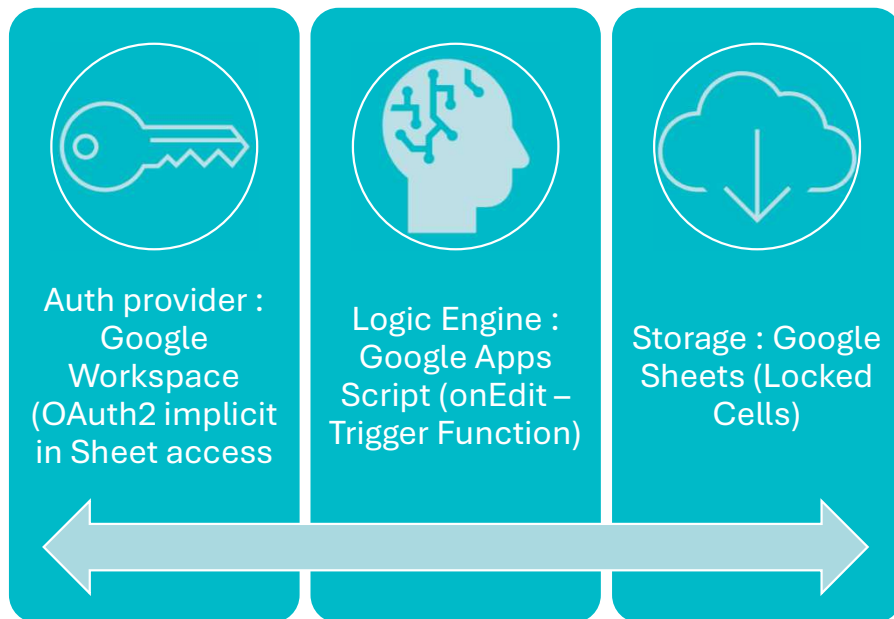
The Identity Architecture	<i>How we leverage Google Workspace to enforce non-repudiable identity tracking.</i>
Timesheet Schema	<i>The validated data structure designed for "outcome-focused" reporting.</i>
Audit Control Framework	<i>The "Detective" and "Preventive" logic gates that protect payroll integrity.</i>
Event Mapping	<i>How governance events (late submissions, edits) are mapped to Slack for operational visibility.</i>



Identity & Access Architecture (The Source of Truth)



Tech Stack :



```
script.js x
1  /**
2   * SAS Timesheet Logic v1
3   * Implements Identity Tracking, Audit Timestamping, and Outcome Validation.
4   */
5  function onEdit(e) {
6      const sheet = e.source.getActiveSheet();
7      const range = e.range;
8      const row = range.getRow();
9      const col = range.getColumn();
10
11     // Scope: Only run on 'Timesheet_Entries' (skip headers)
12     if (sheet.getName() !== "Timesheet_Entries" || row < 2) return;
13
14     const timestamp = new Date();
15     const userEmail = Session.getActiveUser().getEmail();
16
17     // 1. IDENTITY CAPTURE (Preventive Control)
18     // Automatically records who created the row.
19     const emailCell = sheet.getRange(row, 2);
20     if (emailCell.getValue() === "") {
```

```
21         emailCell.setValue(userEmail);
22     }
23
24     // 2. AUDIT TRAIL (Detective Control)
25     // Updates 'Last Edit Timestamp' on every single change.
26     sheet.getRange(row, 9).setValue(timestamp);
27
28     // 3. OUTCOME VALIDATION (Quality Control)
29     // Enforces 10-char min for descriptions to ensure 'delivery-driven' reporting.
30     if (col === 6) {
31         const description = range.getValue();
32         if (description.length > 0 && description.length < 10) {
33             SpreadsheetApp.getUi().alert("GOVERNANCE ERROR: Description must be >10 chars.");
34             range.clearContent(); // Reject input
35             return;
36         }
37     }
38
39     // 4. SUBMISSION LOCK (Governance State)
40     // Locks the time for 'Late Submission' detection.
41     if (col === 7 && range.getValue() === "Submitted") {
42         sheet.getRange(row, 8).setValue(timestamp);
43     }
44 }
```

Timesheet Design & Schema

Outcome-Focused Input: Column F (Description)	<i>Enforces a strict "10-character minimum" via script. Vague entries like "Worked" are auto-rejected to ensure "time-on-site" is replaced with "delivery-output".</i>
Time Normalization: Columns C & D	<i>Dates are validated to ISO week starts. Column E (Hours) is restricted to 0–24 to prevent data entry errors.</i>
State Management	<i>The row status is restricted to a strict state machine: Draft → Submitted → Approved.</i>
Validation	<i>Google Sheets Data Validation (Regex/Length checks) + Apps Script Interceptors.</i>
UI	<i>Protected Ranges (Locking headers and formula columns).</i>



Audit Controls for Payroll (The Governance Layer)



Retroactive editing – System compares last edit timestamp vs submission timestamp, outcome triggers flag for audit.

Late Submission – System checks if submission timestamp is outside of submission week, outcome triggers flag for audit.

Unauthorized changes – system triggers soft lock upon approved trigger, output control breach alert on edit attempts.

Governance Visibility (Slack Event Mapping)



Trigger –
Status
changes to
“submitted”

Payload –
User email
+ total
hours

Submission Event

Trigger –
Logic
violation

Payload –
“User X
edited a
submitted
timesheet
on [Date]”

Audit Flag

Trigger –
Status
changes to
“approved”

Payload –
“Payroll
ready”
signal sent
to finance.

Approval Event

Summary

Week 1 Deliverables :

- ✓ *Template: Identity-aware Google Sheet deployed.*



- ✓ *Governance: Cut-off rules (Sunday 23:59 SAST) defined.*



- ✓ *Audit Logic: Preventive (Validation) and Detective (Flags) controls active.*



Week 2 Goal : Automation

- ✓ *Connecting n8n to the Sheet to read the timestamps we just built.*
- ✓ *Activating the Slack Webhooks for the events mapped on Slide 5.*